

Universal Surgical Alignment (USA)

Multi-Level Video-Language Matching with
Action-Driven Digital Twins

A Comprehensive Master Thesis Proposal & Execution Plan

Grounded in Surgical Data Science, Computer Vision,
and Multimodal Deep Learning

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Part I

Research Foundation

1 Executive Summary

This master thesis proposes **Universal Surgical Alignment (USA)**, a framework that aligns streaming surgical video directly with predefined canonical step checklists using hierarchical video-language pretraining. The system converts raw video into structured **Action-Driven Digital Twins (ActDTs)**—JSON-serialized representations of tool-tissue interactions (Subject-Action-Object-Location-Depth)—and aligns them to multi-level canonical references (Phase → Step → Action → Instrument → Anatomy) in disentangled embedding spaces.

Key Insight

Key Scope Clarification: We treat canonical surgical checklists as predefined procedural references (from clinical protocols or gold-standard dataset annotations), eliminating the preliminary task of generating canonical descriptions from raw transcripts. The entire research focuses on the **core alignment challenge**.

1.1 Core Contributions

1. A **forward-facing video parser** that converts pixel streams into structured ActDT representations
2. **Hierarchical contrastive alignment** across disentangled embedding spaces
3. Robust handling of **skipped/reordered steps** via step-aware padding and temporal rectification
4. **Zero-shot generalization** across surgical specialties without re-annotation

2 Problem Statement

2.1 The Clinical Need

Computer-assisted surgical (CAS) systems require granular, explainable workflow understanding to:

- Monitor clinical procedures in real-time
- Provide safety checkpoints and goal verification
- Alert surgical teams to skipped or misordered steps
- Enable standardized training and quality assurance

2.2 Five Critical Technical Limitations

Table 1: Critical limitations of current surgical AI systems

Limitation	Description	Evidence
Categorical Hard-coding	Traditional models use fixed classification heads, preventing open-vocabulary generalization	Adapting to new surgery requires complete re-annotation
Multi-Center Failure	Models collapse when workflows deviate between hospitals	HecVL: Significant drop from Strasbourg $\rightarrow$ Bern
Semantic Granularity Collapse	Single embedding spaces confuse phase-level and action-level information	HecVL ablations: “Flat” models perform inconsistently
Visual Near-Identity	Consecutive surgical clips are visually almost identical	OR3: CLIP4Clip achieves $<16\%$ R@1
Temporal Blindness	Frame-by-frame models lack global procedural context	SurgPLAN++: Independent classification leads to inconsistent predictions

2.3 The Literature Gap

Literature Gap

No existing paper provides a unified, forward-facing pipeline that:

1. Converts streaming video into structured ActDT representations
2. Aligns these to canonical checklists in real-time
3. Generalizes across surgical specialties without re-training
4. Flags skipped or misordered steps

Table 2: Literature gap analysis

Paper	Strengths	Critical Gaps
HecVL	Hierarchical video-language pretraining	No forward parser; no ActDT; fails on center-specific variations
HCT (MSSU)	Multi-level semantic hierarchy	Fully supervised; massive annotations required; no zero-shot
OR3 / ActDT	Structured digital twin representations	Inverse direction only (text $\rightarrow$ video); no forward parsing
StepAL	Step-aware temporal representations	Frame/clip-level only; no cross-modal alignment
SurgPLAN++	Temporal rectification for online inference	Vision-only; no language grounding; no open-vocabulary

3 Research Questions & Hypothesis

3.1 Primary Research Question

Can we leverage hierarchical video-language alignment to automatically synchronize raw surgical video segments with pre-existing canonical step checklists to enable zero-shot procedural recognition and maintain tracking continuity when steps are skipped or performed out of sequence?

3.2 Sub-Questions

Table 3: Research sub-questions and their solutions

ID	Sub-Question	Addressed By
SQ1	Optimal structured format for video content?	ActDT JSON serialization
SQ2	Maintain alignment when steps skipped?	SFR padding + temporal rectification
SQ3	Transfer to new specialties without retraining?	Open-vocabulary text alignment
SQ4	Metrics to justify alignment quality?	Three-tier evaluation framework

3.3 Thesis Hypothesis

Hypothesis

By converting raw clinical video streams into structured, text-serialized Action-Driven Digital Twins (ActDTs) and performing contrastive learning across separate, hierarchical visual-textual embedding spaces, we can align visually near-identical surgical clips to their correct chronological position on a standard canonical checklist. This alignment remains robust to procedural variations (skips, substitutions) by utilizing step-aware padding and global temporal rectification.

Part II

Technical Methodology

4 System Architecture

4.1 High-Level Overview

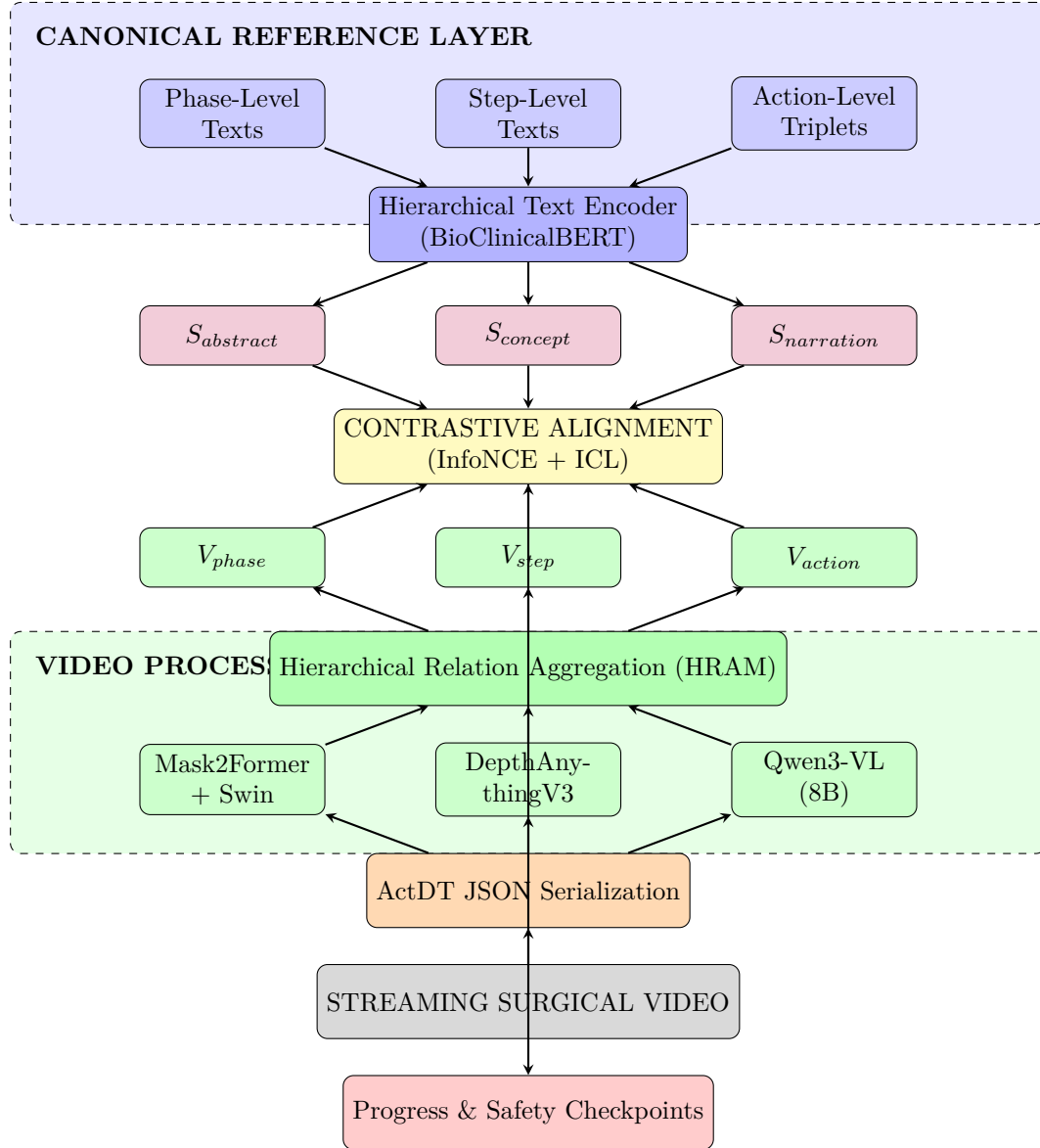


Figure 1: Universal Surgical Alignment (USA) system architecture overview

4.2 Pipeline Stage A: Action-Driven Video Parser

Purpose: Convert high-dimensional pixel streams into structured, text-serialized ActDTs.

4.2.1 A1. Spatio-Temporal Segmentation

- **Model:** Mask2Former with VideoSwin backbone (pretrained on surgical videos)
- **Output:** Pixel-precise segmentation masks for 29+ surgical objects/anatomies
- **Temporal:** VideoSwin processes consecutive frame chunks for motion dynamics

4.2.2 A2. 3D Coordinate Extraction

- **Model:** DepthAnythingV3
- **Output:** Centroid depth for each segmented element
- **Format:** $\mathbb{R}^5$ coordinates $(x_1, y_1, x_2, y_2, z_{\text{depth}})$

4.2.3 A3. Action Attribute Generation

- **Model:** Qwen3-VL-8B (fine-tuned on surgical interactions)
- **Input:** Localized visual regions of interest
- **Output:** Natural language action attributes (tool-tissue interactions)

4.2.4 A4. JSON Serialization

Listing 1: ActDT JSON output format

```

1 {
2   "clip_id": "video_001_clip_042",
3   "interval": [125.0, 130.0],
4   "actions": [
5     {
6       "subject": "Capsulorhexis Forceps",
7       "action": "Tear",
8       "object": "Anterior Capsule",
9       "location_3D": [0.32, 0.45, 0.38, 0.52, 0.67],
10      "confidence": 0.94
11    },
12    {
13      "subject": "Rycroft Cannula",
14      "action": "Inject",
15      "object": "Anterior Chamber",
16      "location_3D": [0.55, 0.48, 0.61, 0.54, 0.45],
17      "confidence": 0.89
18    }
19  ],
20  "predicted_step": "Capsulorhexis",
21  "predicted_phase": "Nucleus Removal Preparation"
22 }
```

4.3 Pipeline Stage B: Spatio-Temporal Hierarchical Aligner

4.3.1 B1. Hierarchical Relation Aggregation Module (HRAM)

- **Function:** Learn semantic relations across task hierarchies
- **Mechanism:** Cross-attention blocks linking Phase $\leftrightarrow$ Step $\leftrightarrow$ Action features
- **Benefit:** Fine-grained tool-action features assist high-level step localization

4.3.2 B2. Disentangled Multi-Space Alignment (HecVL-style)

Table 4: Embedding space configuration

Space	Granularity	Text Type	Visual Aggregation
$S_{\text{narration}}$	Clip-level	Action triplets	Single clip features
S_{concept}	Step-level	Conceptual summaries	Aggregated clip sequences
S_{abstract}	Phase-level	Procedure abstracts	Full phase features

4.3.3 B3. Inter-Task Contrastive Learning (ICL)

Loss Function:

$$\mathcal{L}_{\text{ICL}} = \sum_{(i,j) \in \mathcal{P}} \mathcal{L}_{\text{InfoNCE}}(v_i, t_j) + \lambda \cdot \mathcal{L}_{\text{consistency}}(h_i, h_j) \quad (1)$$

Where:

- $(i, j) \in \mathcal{P} = \text{consistent task pairs (Phase} \leftrightarrow \text{Step, Action} \leftrightarrow \text{Instrument)}$
- $\mathcal{L}_{\text{consistency}}$ = regularization for hierarchical agreement
- λ = weighting hyperparameter

4.4 Pipeline Stage C: Temporal Robustness Mechanisms

4.4.1 C1. Step-Aware Feature Representation (SFR)

Problem: Skipped steps create dimensional inconsistency.

Solution:

$$\mathbf{f}_{S_k} = \begin{cases} \frac{1}{|C_k|} \sum_{c \in C_k} \mathbf{f}_c & \text{if } |C_k| > 0 \\ \frac{1}{|V|} \sum_{c \in V} \mathbf{f}_c & \text{if step skipped} \end{cases} \quad (2)$$

4.4.2 C2. Temporal Rectification (SurgPLAN++)

Problem: Online streaming lacks global context.

Solution:

1. **Mirroring:** Extend partial video by reversing existing content
2. **Center-Duplication:** Replicate middle segments to simulate completion
3. **Dynamic Rectification:** Continuously refine predictions as new frames arrive

$$R_{t+1} = \text{Rectify}(R_t, P_{\text{global}}(V_{0:t+1})) \quad (3)$$

5 Cross-Procedure Generalization

5.1 Open-Vocabulary Checklist Substitution

Key Insight

The neural network never changes; only the text input file changes.

Table 5: Cross-procedure vocabulary mapping

Surgery Type	Phase Example	Step Example	Instrument
Cataract	Nucleus Removal	Capsulorhexis	Phaco Handpiece
Cholecystectomy	Calot Triangle Dissection	Clipping Cystic Artery	Clip Applier
Prostatectomy	Bladder Neck Transection	Posterior Dissection	Robotic Grasper
Gastric Bypass	Gastric Pouch Creation	Pouch Stapling	Linear Stapler

5.2 Transfer Mechanism

Algorithm 1 Cross-Procedure Transfer Protocol

```

1: function TRANSFERTONNEWSURGERY(model, new_checklist_path)
2:     ▷ No retraining required - just load new canonical texts
3:     new_canonical ← LOADJSON(new_checklist_path)
4:
5:     ▷ Encode new texts in same embedding spaces
6:     phase_emb ← model.text_encoder(new_canonical['phases'], space='abstract')
7:     step_emb ← model.text_encoder(new_canonical['steps'], space='concept')
8:     action_emb ← model.text_encoder(new_canonical['actions'], space='narration')
9:
10:    return model.with_canonical(phase_emb, step_emb, action_emb)
11: end function

```

Part III

Evaluation Framework

6 Three-Tier Evaluation Suite

6.1 Tier 1: Zero-Shot Recognition Benchmarks

Table 6: Tier 1 evaluation metrics

Metric	Description	Target
Frame Accuracy	Correct frame classifications / Total frames	> 85%
Step Accuracy	Correct step predictions / Total steps	> 75%
Phase F1-Score	Harmonic mean of precision and recall	> 80%
Video Jaccard (IoU)	Temporal overlap of predicted vs. GT segments	> 70%
Recall@K	% correct retrievals in top K results	R@1 > 40%

6.2 Tier 2: Alignment Quality & Process Integrity

Alignment Discrepancy Index (ADI):

$$ADI = \frac{1}{M} \sum_{i=1}^M \left(|t_{\text{start},i}^{\text{GT}} - t_{\text{start},i}^{\text{Pred}}| + |t_{\text{end},i}^{\text{GT}} - t_{\text{end},i}^{\text{Pred}}| \right) \quad (4)$$

Table 7: Tier 2 evaluation metrics

Metric	Description	Relevance
ADI	Average temporal offset (Eq. 4)	Temporal precision
SOI	Kendall’s τ with canonical order	Chronological correctness
HCS	% frames where Phase-Step-Action agree	Multi-level coherence
CDR	Correctly flagged milestones / Total	Safety verification

6.3 Tier 3: Anomaly & Cross-Center Robustness

Table 8: Tier 3 evaluation metrics

Metric	Formula/Description	Purpose
SDR	Correctly flagged skips / Actual skips	Safety alerting
FAR	Incorrect skip flags / Total flags	Avoiding alert fatigue
Cross-Center Drop	$(Acc_{\text{train}} - Acc_{\text{new}}) / Acc_{\text{train}}$	Generalization
Zero-Shot Acc	Performance on unseen surgery types	Deployment flexibility

6.4 Novel Composite Metric: Procedural Alignment Score (PAS)

$$\text{PAS} = 0.3 \times \text{Phase_Acc} + 0.4 \times \text{Step_Acc} + 0.3 \times \text{SDR} \quad (5)$$

7 Experimental Design

7.1 Experiment Matrix

Table 9: Complete experiment matrix

ID	Experiment	Datasets	Primary Metrics
E1	Supervised baseline	CATARACTS	Phase Acc, Step Acc
E2	Zero-shot: Cat $\rightarrow$ Chole	CAT-SG $\rightarrow$ Cholec80	Zero-Shot Acc, PAS
E3	Zero-shot: Cat $\rightarrow$ Prost	CAT-SG $\rightarrow$ PSI-AVA	Zero-Shot Acc, PAS
E4	Cross-center robustness	Stras $\leftrightarrow$ Bern	Drop Rate, SOI
E5	Ablation: Single vs. hier.	CAT-SG	Phase Acc, HCS
E6	Ablation: $\pm$ ICL loss	CAT-SG	Step Acc, HCS
E7	Ablation: $\pm$ SFR padding	Cataract-1k	SDR, FAR
E8	Ablation: $\pm$ rectification	StrasBypass70	SOI, ADI
E9	Skip detection stress test	Synthetic + Real	SDR, FAR, PAS
E10	Latency benchmark	All	ms/frame, FPS

7.2 Baseline Comparisons

Table 10: Baseline methods for comparison

Method	Type	What It Represents
Trans-SVNet	Supervised	Current SOTA surgical phase recognition
CLIP4Clip	Zero-shot	Standard video-language retrieval
HecVL (reproduced)	Zero-shot	Hierarchical VL without ActDT
Ours (USA)	Zero-shot	Full proposed framework

Part IV

Data Requirements

8 Dataset Inventory

8.1 Primary Datasets (Required)

Table 11: Primary datasets required for the project

Dataset	Surgery	Annotations	Size	Role
CAT-SG / CaDIS	Cataract	29 classes, 1.8M relations, scene graphs	56 videos	Primary benchmark
CATARACTS	Cataract	19 steps, tool annotations	50 videos	Step recognition
Cataract-1k	Cataract	13 chronological steps	1000 videos	SFR evaluation
Cholec80	Cholecystectomy	7 phases, tool presence	80 videos	Cross-procedure
PSI-AVA	Prostatectomy	11 phases, 21 steps, 16 actions, 7 instruments	Multi-center	Cross-specialty
StrasBypass70 / BernBypass70	Gastric Bypass	Phase/step labels	140 videos	Multi-center

8.2 Pretraining Datasets

Table 12: Pretraining datasets

Dataset	Content	Size	Purpose
SurgVLP / SVL-25K	Multi-procedure lectures	25K clips	Encoder pretraining
MedVideoCap-55K	Diverse clinical clips	55K videos	Generalization
HowTo100M (subset)	Instructional videos	Filtered	VL foundations

8.3 Artifacts to Create

Table 13: Artifacts to be created during the project

Artifact	Description	Effort	Priority
Canonical JSON templates	MSSU hierarchy per surgery	Medium	HIGH
Cross-procedure vocabulary	Standardized ontology	High	HIGH
Evaluation splits	Zero-shot vs. fine-tuned	Low	MEDIUM
Synthetic skip sequences	Artificial skip scenarios	Medium	MEDIUM

Part V

Implementation Roadmap

9 Master Timeline (12 Months)

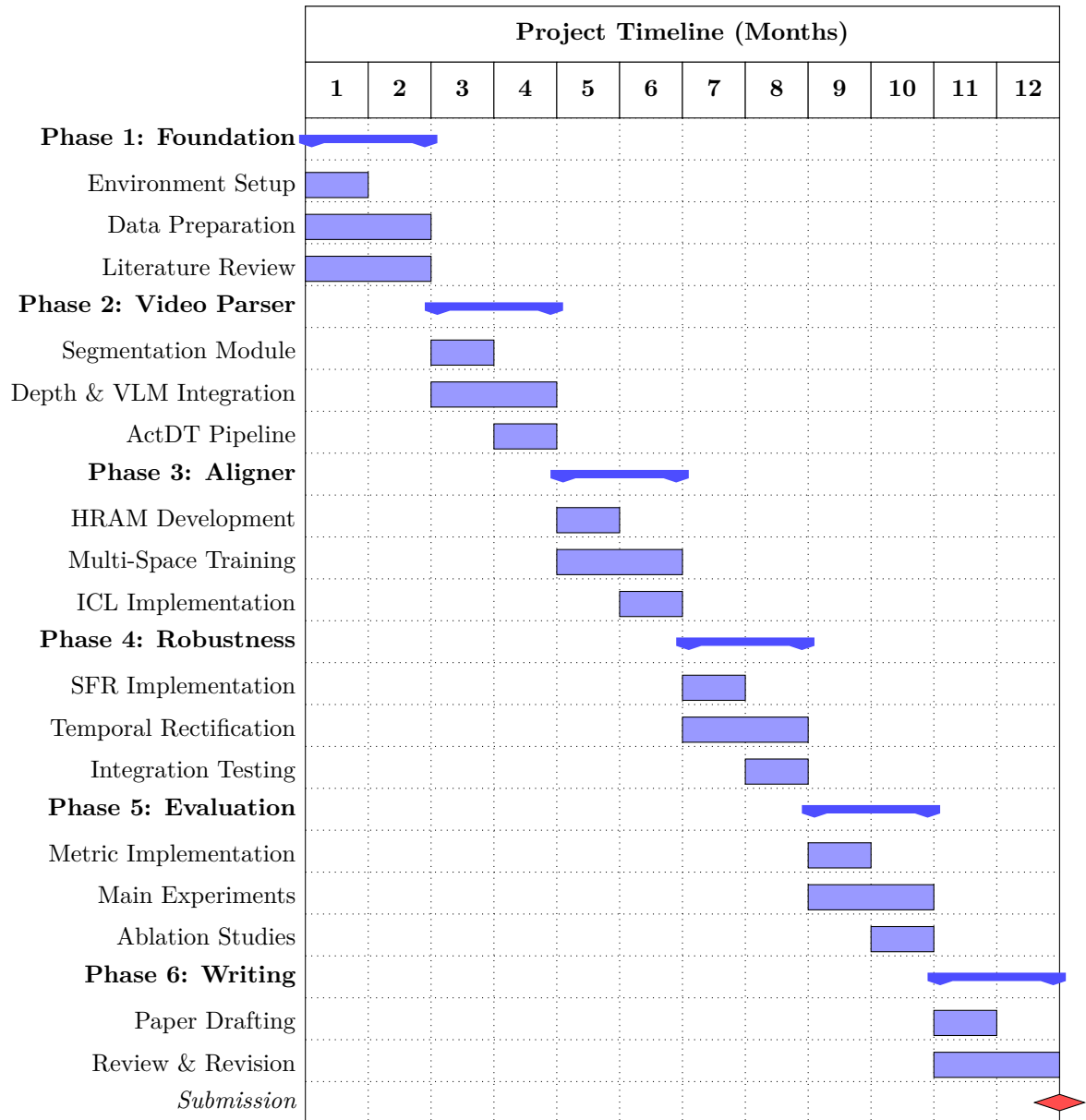


Figure 2: 12-month project Gantt chart

10 Detailed Task Breakdown

10.1 Phase 1: Foundation (Months 1-2)

10.1.1 Development Tasks

Table 14: Phase 1 Development Tasks

ID	Task	Priority	Hours	Deps
D1.1	Set up dev environment (PyTorch, CUDA, W&B)	HIGH	4	–
D1.2	Configure multi-GPU infrastructure	HIGH	8	D1.1
D1.3	Download CATARACTS dataset	HIGH	4	D1.1
D1.4	Download CAT-SG dataset	HIGH	4	D1.1
D1.5	Download Cholec80 dataset	HIGH	4	D1.1
D1.6	Download PSI-AVA dataset	MEDIUM	6	D1.1
D1.7	Download StrasBypass70/BernBypass70	MEDIUM	6	D1.1
D1.8	Preprocess videos to 1fps	HIGH	12	D1.3–7
D1.9	Download pretrained backbones	HIGH	4	D1.1
D1.10	Download Qwen3-VL-8B	HIGH	4	D1.1
D1.11	Set up SurgVLP pretraining pipeline	MEDIUM	16	D1.1
D1.12	Create unified data loading framework	HIGH	16	D1.8
D1.13	Implement evaluation harness	MEDIUM	12	D1.1
D1.14	Set up experiment tracking	LOW	4	D1.1

Phase 1 Development Total: 104 hours

10.1.2 Writing Tasks

Table 15: Phase 1 Writing Tasks

ID	Task	Priority	Hours	Deps
W1.1	Literature review outline	HIGH	6	–
W1.2	Deep-dive: HecVL	HIGH	8	–
W1.3	Deep-dive: HCT (MSSU)	HIGH	8	–
W1.4	Deep-dive: OR3 / ActDT	HIGH	8	–
W1.5	Deep-dive: StepAL	HIGH	6	–
W1.6	Deep-dive: SurgPLAN++	HIGH	6	–
W1.7	Deep-dive: Dynamic Scene Graphs	MEDIUM	4	–
W1.8	Draft Related Work v1	MEDIUM	12	W1.2–7
W1.9	Create problem statement diagram	MEDIUM	4	W1.8

Phase 1 Writing Total: 62 hours

10.2 Phase 2: Action-Driven Video Parser (Months 3-4)

Phase 2 Development Total: 180 hours

10.3 Phase 3: Hierarchical Aligner (Months 5-6)

Phase 3 Development Total: 248 hours

10.4 Phase 4: Temporal Robustness (Months 7-8)

Phase 4 Development Total: 148 hours

10.5 Phase 5: Evaluation (Months 9-10)

Phase 5 Development Total: 178 hours

Table 16: Phase 2 Key Development Tasks

ID	Task	Priority	Hours	Deps
D2.1	Implement Mask2Former module	HIGH	20	D1.9
D2.2	Fine-tune on CaDIS classes	HIGH	24	D2.1
D2.3	Integrate VideoSwin backbone	HIGH	16	D2.1
D2.4	DepthAnythingV3 integration	HIGH	12	D1.1
D2.7	Qwen3-VL action generator	HIGH	16	D1.10
D2.10	JSON serialization pipeline	HIGH	8	D2.9
D2.11	End-to-end video-to-ActDT	HIGH	16	All
D2.14	Generate ActDTs for all videos	HIGH	20	D2.11

Table 17: Phase 3 Key Development Tasks

ID	Task	Priority	Hours	Deps
D3.5	Cross-procedure ontology	HIGH	16	D3.1–4
D3.9	HARAM cross-attention module	HIGH	20	D3.8
D3.11	ICL loss implementation	HIGH	12	D3.10
D3.14	Train encoders on SurgVLP	HIGH	40	D3.12
D3.15	Fine-tune on CAT-SG	HIGH	32	D3.14
D3.16	Hyperparameter tuning	MEDIUM	24	D3.15

Table 18: Phase 4 Key Development Tasks

ID	Task	Priority	Hours	Deps
D4.4	SFR-enhanced feature module	HIGH	12	D4.3
D4.8	SurgPLAN++ rectification loop	HIGH	20	D4.5–7
D4.11	Skip detection alerting	HIGH	8	D4.4
D4.13	Integrate all robustness modules	HIGH	16	D4.4, D4.9

Table 19: Phase 5 Experiment Tasks

ID	Experiment	Priority	Hours	Deps
D5.6	E1: Supervised baseline	HIGH	8	D3.15
D5.7	E2: Zero-shot Cat→Chole	HIGH	12	D5.5
D5.8	E3: Zero-shot Cat→Prost	HIGH	12	D5.5
D5.9	E4: Cross-center evaluation	HIGH	16	D5.5
D5.10–13	E5–E8: Ablation studies	HIGH	32	D5.5
D5.14	E9: Skip detection stress test	HIGH	12	D4.10

10.6 Phase 6: Paper Writing (Months 11-12)

Phase 6 Writing Total: 160 hours

11 Summary Statistics

Table 20: Total effort by phase

Phase	Development	Writing	Total
Phase 1: Foundation	104 hrs	62 hrs	166 hrs
Phase 2: Video Parser	180 hrs	22 hrs	202 hrs
Phase 3: Hierarchical Aligner	248 hrs	34 hrs	282 hrs
Phase 4: Robustness	148 hrs	24 hrs	172 hrs
Phase 5: Evaluation	178 hrs	70 hrs	248 hrs
Phase 6: Writing	—	160 hrs	160 hrs
TOTAL	858 hrs	372 hrs	1,230 hrs

Weekly Commitment

- 12 months $\times$ 4 weeks = 48 weeks
- 1,230 hours / 48 weeks $\approx$ **25.6 hours/week**

12 Critical Path Items

Critical Path Item

Highest Priority Development (Blocks Everything):

1. **D2.11:** End-to-end video-to-ActDT pipeline — Core technical contribution
2. **D3.5:** Cross-procedure instrument ontology — Blocks cross-surgery experiments
3. **D3.11:** ICL loss implementation — Key to hierarchical consistency
4. **D4.4:** SFR-enhanced feature module — Key to skip-handling claims
5. **D5.7–D5.9:** Zero-shot transfer experiments — Main paper claims

Critical Path Item

Highest Priority Writing (Blocks Submission):

1. **W1.2–W1.6:** Paper deep-dives — Required before methodology design
2. **W3.3:** Loss function documentation — Reviewer scrutiny area
3. **W5.5:** Results analysis — Makes or breaks the paper
4. **W6.2:** Introduction — Sets entire paper narrative
5. **W6.12:** Advisor review — Catches major issues before submission

13 Risk Mitigation

Table 21: Risk assessment and mitigation strategies

Risk	Likelihood	Impact	Mitigation
Dataset access delays	Medium	High	Start requests immediately; identify backups
GPU resource constraints	Medium	High	Reserve cloud credits; optimize batch sizes
Qwen3-VL integration	Medium	Medium	Have BLIP-2 as backup VLM
Cross-center data unavailable	Low	High	Use synthetic variations
Performance below baselines	Low	High	Extensive ablations; adjust architecture

14 Target Venues

Table 22: Target publication venues

Venue	Deadline	Fit	Notes
MICCAI 2025	~March 2025	*****	Primary target
IPCAI 2025	~December 2024	***	Surgical AI focus
CVPR 2025	~November 2024	***	Broader audience
Medical Image Analysis	Rolling	*****	Extended version
IEEE TMI	Rolling	***	Strong MI journal

15 Immediate Next Actions

15.1 This Week

- ☐ Set up Git repository with project structure
- ☐ Begin HecVL paper deep-read (W1.2)
- ☐ Submit dataset access requests (CATARACTS, CAT-SG, PSI-AVA)
- ☐ Configure development environment (D1.1)

15.2 Next Week

- ☐ Complete HecVL notes
- ☐ Begin OR3 paper deep-read (W1.4)
- ☐ Download and verify first dataset
- ☐ Draft MSSU hierarchy for cataract surgery

15.3 Month 1 Milestone

- ☐ All paper deep-dives complete
- ☐ Development environment fully configured
- ☐ At least 3 datasets downloaded and verified
- ☐ Related Work section v1 drafted

Appendices

A Canonical Hierarchy Examples

A.1 Cataract Surgery

Listing 2: Cataract surgery canonical hierarchy

```

1 {
2   "procedure": "Cataract Surgery",
3   "phases": [
4     {
5       "id": "P1",
6       "name": "Preparation",
7       "steps": [
8         {"id": "S1", "name": "Sterile Draping",
9          "actions": ["Drape", "Position"]},
10        {"id": "S2", "name": "Anesthesia",
11         "actions": ["Inject", "Apply"]}
12      ]
13    },
14    {
15      "id": "P2",
16      "name": "Incision",
17      "steps": [
18        {"id": "S3", "name": "Main Incision",
19         "actions": ["Cut"],
20         "instruments": ["Primary Knife"],
21         "anatomy": ["Cornea"]},
22        {"id": "S4", "name": "Side Port",
23         "actions": ["Cut"],
24         "instruments": ["Secondary Knife"],
25         "anatomy": ["Cornea"]}
26      ]
27    },
28    {
29      "id": "P3",
30      "name": "Capsulorhexis",
31      "steps": [
32        {"id": "S5", "name": "Viscoelastic Injection",
33         "actions": ["Inject"],
34         "instruments": ["Rycroft Cannula"],
35         "anatomy": ["Anterior Chamber"]},
36        {"id": "S6", "name": "Capsule Tearing",
37         "actions": ["Tear", "Grasp"],
38         "instruments": ["Capsulorhexis Forceps"],
39         "anatomy": ["Anterior Capsule"]}
40      ]
41    }
42  ]
43 }

```

A.2 Cholecystectomy

Listing 3: Cholecystectomy canonical hierarchy

```

1 {
2   "procedure": "Laparoscopic Cholecystectomy",
3   "phases": [
4     {
5       "id": "P1",
6       "name": "Preparation",
7       "steps": [
8         {"id": "S1", "name": "Trocarn Placement",
9          "actions": ["Insert", "Inflate"]}
10      ]
11    },
12    {
13      "id": "P2",
14      "name": "Calot Triangle Dissection",
15      "steps": [
16        {"id": "S2", "name": "Cystic Duct Identification",
17         "actions": ["Dissect", "Expose"],
18         "instruments": ["Grasper", "Hook"],
19         "anatomy": ["Cystic Duct"]},
20        {"id": "S3", "name": "Cystic Artery Identification",
21         "actions": ["Dissect", "Expose"],
22         "instruments": ["Grasper", "Hook"],
23         "anatomy": ["Cystic Artery"]}
24      ]
25    },
26    {
27      "id": "P3",
28      "name": "Clipping and Cutting",
29      "steps": [
30        {"id": "S4", "name": "Clip Cystic Duct",
31         "actions": ["Clip"],
32         "instruments": ["Clip Applier"],
33         "anatomy": ["Cystic Duct"]},
34        {"id": "S5", "name": "Clip Cystic Artery",
35         "actions": ["Clip"],
36         "instruments": ["Clip Applier"],
37         "anatomy": ["Cystic Artery"]},
38        {"id": "S6", "name": "Cut Cystic Duct",
39         "actions": ["Cut"],
40         "instruments": ["Scissors"],
41         "anatomy": ["Cystic Duct"]}
42      ]
43    }
44  ]
45 }

```

B Evaluation Metric Implementations

Listing 4: Evaluation metric implementations

```

1 import numpy as np
2 from scipy.stats import kendalltau
3
4 def alignment_discrepancy_index(gt_boundaries, pred_boundaries):
5     """

```

```

6     ADI: Average temporal offset between GT and predicted
7     step boundaries.
8
9     Args:
10        gt_boundaries: List of (start, end) tuples for GT
11        pred_boundaries: List of (start, end) tuples for preds
12
13     Returns:
14        ADI score (lower is better)
15        """
16     assert len(gt_boundaries) == len(pred_boundaries)
17
18     total_discrepancy = 0
19     for (gt_s, gt_e), (pred_s, pred_e) in zip(
20         gt_boundaries, pred_boundaries):
21         total_discrepancy += abs(gt_s - pred_s) + abs(gt_e - pred_e)
22
23     return total_discrepancy / len(gt_boundaries)
24
25
26 def sequence_order_integrity(canonical_order, predicted_order):
27     """
28     SOI: Kendall's tau correlation between canonical and
29     predicted step orders.
30
31     Args:
32        canonical_order: List of step IDs in canonical order
33        predicted_order: List of step IDs in predicted order
34
35     Returns:
36        Kendall's tau coefficient (-1 to 1, higher is better)
37        """
38     canonical_ranks = {step: i for i, step in
39                        enumerate(canonical_order)}
40     pred_ranks = [canonical_ranks.get(step, len(canonical_order))
41                  for step in predicted_order]
42
43     tau, _ = kendalltau(list(range(len(predicted_order))),
44                        pred_ranks)
45     return tau
46
47
48 def hierarchical_consistency_score(phase_preds, step_preds,
49                                   action_preds, hierarchy_map):
50     """
51     HCS: Percentage of frames where Phase-Step-Action
52     predictions are consistent.
53
54     Args:
55        phase_preds: Array of phase predictions per frame
56        step_preds: Array of step predictions per frame
57        action_preds: Array of action predictions per frame
58        hierarchy_map: Dict mapping valid relationships
59
60     Returns:
61        HCS percentage (0-100, higher is better)
62        """
63     consistent_frames = 0

```

```

64     total_frames = len(phase_preds)
65
66     for phase, step, action in zip(phase_preds, step_preds,
67                                   action_preds):
68         phase_step_ok = step in hierarchy_map[
69             'phase_to_steps'].get(phase, [])
70         step_action_ok = action in hierarchy_map[
71             'step_to_actions'].get(step, [])
72
73         if phase_step_ok and step_action_ok:
74             consistent_frames += 1
75
76     return (consistent_frames / total_frames) * 100
77
78
79 def skip_detection_metrics(gt_skips, pred_skips, total_flags):
80     """
81     SDR and FAR: Skip detection recall and false alarm rate.
82
83     Args:
84         gt_skips: Set of actually skipped step IDs
85         pred_skips: Set of predicted skipped step IDs
86         total_flags: Total number of skip flags raised
87
88     Returns:
89         (SDR, FAR) tuple
90     """
91     true_positives = len(gt_skips & pred_skips)
92     false_positives = len(pred_skips - gt_skips)
93
94     sdr = true_positives / len(gt_skips) if gt_skips else 1.0
95     far = false_positives / total_flags if total_flags else 0.0
96
97     return sdr, far
98
99
100 def procedural_alignment_score(phase_acc, step_acc, sdr):
101     """
102     PAS: Composite metric for overall alignment quality.
103
104     Args:
105         phase_acc: Phase accuracy (0-1)
106         step_acc: Step accuracy (0-1)
107         sdr: Skip detection recall (0-1)
108
109     Returns:
110         PAS score (0-1, higher is better)
111     """
112     return 0.3 * phase_acc + 0.4 * step_acc + 0.3 * sdr

```

C Key Paper Reference Summary

D Loss Function Formulations

Table 23: Key paper reference summary

Paper	Key Technique	What to Extract	Gap It Leaves
HecVL	Hierarchical VL pretraining	Multi-space contrastive learning	No forward parser; no ActDT
HCT (MSSU)	Multi-level semantic understanding	HRAM cross-attention; ICL loss	Fully supervised; no zero-shot
OR3	Action-driven digital twins	ActDT JSON schema	Inverse direction only
StepAL	Step-aware feature representation	SFR padding mechanism	No cross-modal alignment
SurgPLAN++	Temporal phase localization	Mirroring; rectification loop	Vision-only; no language
CAT-SG	Dynamic scene graphs	Graph structure; 29-class ontology	No VL alignment

D.1 InfoNCE Contrastive Loss

The base contrastive loss for aligning visual and textual representations:

$$\mathcal{L}_{\text{InfoNCE}} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(\text{sim}(v_i, t_i)/\tau)}{\sum_{j=1}^N \exp(\text{sim}(v_i, t_j)/\tau)} \quad (6)$$

where $\text{sim}(\cdot, \cdot)$ is cosine similarity and τ is the temperature parameter.

D.2 Hierarchical Consistency Loss

To enforce agreement across hierarchical levels:

$$\mathcal{L}_{\text{consistency}} = \sum_{l=1}^{L-1} \text{KL} \left(p^{(l)} \| p_{\text{agg}}^{(l+1)} \right) \quad (7)$$

where $p^{(l)}$ is the prediction distribution at level l and $p_{\text{agg}}^{(l+1)}$ is the aggregated distribution from the finer level.

D.3 Complete Training Objective

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{InfoNCE}}^{\text{narr}} + \mathcal{L}_{\text{InfoNCE}}^{\text{concept}} + \mathcal{L}_{\text{InfoNCE}}^{\text{abstract}} + \lambda_1 \mathcal{L}_{\text{ICL}} + \lambda_2 \mathcal{L}_{\text{consistency}} \quad (8)$$

E Hardware Requirements

E.1 Estimated Training Times

Table 24: Recommended hardware specifications

Component	Minimum	Recommended
GPU	1× NVIDIA A100 40GB	4× NVIDIA A100 80GB
CPU	32 cores	64 cores
RAM	128 GB	256 GB
Storage	2 TB SSD	8 TB NVMe SSD
Network	10 Gbps	100 Gbps (for distributed)

Table 25: Estimated training times per experiment

Stage	1× A100	4× A100
SurgVLP pretraining	72 hours	20 hours
CAT-SG fine-tuning	24 hours	8 hours
Full experiment suite	120 hours	40 hours

References

Note: References will be added using BibTeX. Key citations include:

- [1] HecVL: Hierarchical Video-Language Pretraining for Zero-shot Surgical Phase Recognition
- [2] HCT: Hierarchical Context Transformer for Multi-level Semantic Scene Understanding
- [3] OR3: Reasoning Text-to-Video Retrieval for Operating Room Clips via Action-Driven Digital Twins
- [4] StepAL: Step-aware Active Learning for Cataract Surgical Videos
- [5] SurgPLAN++: Universal Surgical Phase Localization Network
- [6] CAT-SG: Dynamic Scene Graph Representation for Surgical Video
- [7] Mask2Former: Masked-attention Mask Transformer for Universal Image Segmentation
- [8] VideoSwin: Video Swin Transformer
- [9] DepthAnything: Unleashing the Power of Large-Scale Unlabeled Data
- [10] Qwen-VL: A Versatile Vision-Language Model